



Ethics in publishing: From plagiarism to artificial intelligence

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ABSTRACT

The digitization of research has transformed how evidence is gathered, hypotheses are generated, and manuscripts are written, introducing ethical challenges related to plagiarism, authorship, and artificial intelligence (AI) in medical writing. This structured narrative review, informed by a comprehensive database search (PubMed/MEDLINE, EMBASE, Scopus, and Web of Science; January 2005–March 2026), examines contemporary approaches to prevent plagiarism and ensure the ethical use of AI in medical publishing. While a systematic search of PubMed, EMBASE, Scopus, and Web of Science was applied, the objective was conceptual synthesis rather than quantitative meta-analysis. AI can enhance efficiency and quality, its misuse, through unacknowledged use, over-reliance, or biased outputs, poses a threat to scholarly integrity. Safeguarding trust in medical literature requires a proactive framework that combines plagiarism detection, mandatory AI disclosure, ethical training, and strict editorial oversight. Ultimately, technology may support, but cannot replace, the accountability and ethical responsibility of human authors.

Take-home messages

1. AI can assist with medical writing, but it must be used transparently and under human oversight to preserve accuracy, originality, and accountability.

2. Preventing plagiarism and ensuring ethical authorship remain central to maintaining trust in scientific publishing in the digital era.

1. Introduction

With the increasing digitization of research, new ethical dilemmas are appearing [1]. To find evidence, produce testable hypotheses, and compose scholarly articles, today's researchers mostly rely on internet databases, software tools, and digital platforms [2]. While these tools offer many benefits, they also raise questions about integrity and transparency [3].

One key area is the way researchers search for and select studies. Using reputable databases and applying clear search strategies helps

reduce bias and improve the quality of evidence synthesis [4]. Likewise, it is important to use digital technologies to generate hypotheses carefully, making sure that the ideas are based on current information rather than just algorithms [5].

Language use is another ethical concern. Researchers are expected to write clearly and accurately in English, avoiding "tortured phrases that distort meaning" [6]. It's important to paraphrase properly to prevent plagiarism and maintain the source's original meaning [7].

Authorship and contribution are also under greater scrutiny [8]. In increasingly digital and collaborative processes, it's critical to specify exactly who did what and to adhere to established authorship standards [9,10]. Researchers must also disclose any conflicts of interest (COI) and provide data availability statements for transparency [11].

Despite these long-standing principles, plagiarism—whether of ideas, text, or graphics—remains a major concern undermining trust [12]. While software detects textual plagiarism, images often require manual review. The ease of online access and proliferation of journals has facilitated copying and redundant publication, prompting global

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Form of Plagiarism	Definition / Example
Ideas	Uncredited use of others' unpublished hypotheses, methods, or concepts.
Verbatim (Direct) Copying	Word-for-word copying of text (with or without citation).
Paraphrasing	Copying text but changing words/synonyms or order to obscure overlap.
Self-plagiarism (Recycling)	Reusing one's own published text/data (e.g. duplicate publication) without disclosure.
Translational Plagiarism	Republishing the same work in another language without permission.
Graphics Plagiarism	Using figures, tables, or images from others without permission.
Citation Manipulation	Falsely citing sources or "padding" references to hide copied content.
Compound Plagiarism	Multiple forms (e.g. text + image), often involving paid editing agencies.

Fig. 1. Common instances of plagiarism (adapted from Zimba & Gasparyan, 2021 [12]).

bodies (COPE, CSE, WAME, ICMJE) to enforce stricter definitions and demand full author accountability [12,13].

However, new technologies introduce fresh challenges. Large language models (e.g., ChatGPT) and artificial intelligence (AI) tools can greatly enhance writing efficiency, but they also raise integrity concerns [14]. Many editors worry AI may facilitate plagiarism: indeed, science publishers have already banned unacknowledged AI-generated text due to “unoriginality” risks [15]. The rise of AI, digital libraries, and social media means authors and editors face unprecedented scenarios—some of which lack clear guidelines [13,15]. For instance, Alnaimat et al. argue that ethical norms (authorship, COI, data sharing, etc.) must be updated to address AI’s influence on reporting standards [3,13].

To eliminate uncertainty, we differentiate between AI tools (software systems utilizing machine learning or huge language models) [16], AI-generated content (text or pictures autonomously created by AI) [17], and AI-assisted writing (human-directed composition with AI help under author supervision) [18]. These distinctions are essential because ethical responsibility and disclosure requirements depend on the degree of human control.

2. Aim

To address this gap, this study aims to provide a comprehensive overview of innovative methods for preventing plagiarism and ensuring the ethical integration of AI in medical writing. The review identifies best practices that preserve scientific integrity and promote confidence in the digital age of research by examining evolving ethical frameworks, technical safeguards, and editorial standards.

3. Search strategy

This review was designed as a structured narrative review rather than a systematic review or meta-analysis. We followed the suggested guidelines and carried out a thorough literature search [16]. The search covered publications from January 2005 to December 2025 and was limited to English-language articles. MEDLINE/PubMed, EMBASE, Scopus, and Web of Science were among the databases we searched using controlled vocabulary and terms pertaining to “publication

ethics,” “plagiarism detection,” “artificial intelligence in writing,” and “medical journal guidelines.” The inclusion criteria covered articles that discuss plagiarism and AI in the context of publishing. We also looked at the editing standards and procedures of ICMJE, COPE, CSE, and WAME journals. We documented all databases searched, keywords and MeSH terms used, date ranges, and inclusion/exclusion criteria as part of our clear and repeatable search methodology [19].

4. Detection and prevention of plagiarism

Plagiarism is the use of someone else’s ideas, words, or visuals without proper attribution [12,20,21]. Zimba and Gasparyan highlight practical approaches to preventing plagiarism and promoting academic integrity [12]. They distinguish plagiarism of ideas, texts, and images, noting that graphic plagiarism requires special attention since software often misses it. Fig. 1 summarizes common types of plagiarism.

4.1. Tools for detection

Several tools help journals check for plagiarism using text-matching software, such as iThenticate, to scan submissions for similarity [22]. With its ability to provide both source-specific matches and general similarity, iThenticate is the most sophisticated anti-plagiarism technology [12,23]. Editors should configure filters (e.g., excluding quoted or reference text) to reduce false alarms [12]. High similarity scores (>30–35%) generally trigger editorial scrutiny or rejection [24]. In parallel, AI-oriented content detectors (e.g., GPTZero, Turnitin AI Detection, CrossPlag AI Detector, Sapling AI Detector, and Copyleaks) have been introduced to flag potentially AI-generated text [25,26]. However, reported accuracy varies across contexts, and both false positives and false negatives occur; these tools should be used cautiously and cannot replace human editorial judgment [26,27].

Although these technologies can identify exact copying and recycled text, they are unable to locate inappropriate concept reuse or altered images [28]. Image plagiarism poses greater challenges: Google’s reverse image search may detect exact duplicates, but altered or semantically similar graphics often escape detection [29]. For this reason, journal editors are advised to instruct authors on what

constitutes image plagiarism and require permission for any reused artwork [30].

Emerging AI-driven image verification tools, such as Hive Moderation, Microsoft Image Match, and Google Lens visual matching, attempt to identify duplicated or manipulated graphics using computer vision algorithms [31,32]. Although these methods improve screening capabilities, they are still constrained in identifying semantically changed or partly transformed images. Consequently, expert editorial review and clear author declarations remain essential safeguards [33].

4.2. Prevention tactics

Educating authors is vital for preventing unintentional plagiarism, with training focused on effective literature searching, critical source verification to avoid duplicating prior work, and the development of academic English writing skills [12]. Good paraphrasing is key: authors should understand the main ideas and then write them in their own words [34]. If they use exact words from a source, they must use quotation marks and give credit [35,36]. Even when ideas are rephrased, the source should still be mentioned [35,36].

Technology-based tools in writing articles for publication, like Grammarly, can help editors to write more clearly, but they cannot replace knowing the plagiarism guidelines [37,38]. Authors, who are also weak in English skills and have little writing experience, may push some beginners to copy text [39].

By combining software checks with human judgment, careful reading of flagged overlaps and scrutiny of citations, editors can catch most misconduct [12]. We urge journals to offer or endorse author training on writing ethics, and to cite COPE/CSE definitions of plagiarism in their instructions [12,40].

5. AI for drafting and AI-based research

The drafting process is greatly aided by AI tools that are excellent at expanding text, giving predictive text capabilities, and delivering autocompletion functions [41]. For example, ChatGPT and other AI technologies may significantly enhance the productivity and quality of authoring scientific review articles. This is particularly useful for developing writing styles, expanding content, and making outlines [42]. Large language models (LLMs) like ChatGPT can generate fluent prose, and AI-driven design tools (e.g., Canva's templates) can help authors produce figures quickly [43,44]. These innovations offer efficiency and may improve initial drafts, but they also demand caution [44]. Experts unanimously stress that AI-generated content requires strict human oversight, because it could produce material that deviates from logical coherence or factual veracity [45,46]. AI tools may produce authoritative-sounding text that is incorrect, incomplete, or biased if unchecked [9].

Notably, leading publishers such as Springer Nature do not allow ChatGPT to be listed as a co-author. Instead, they require that any use of AI tools be transparently reported in the Methods section or the Acknowledgments, since the tool itself cannot take responsibility for the work [47]. Academic Medicine similarly urges transparent reporting of AI use and awareness of its limitations [48]. Despite highlighting ChatGPT's potential, a recent systematic study in healthcare scholarship cautions against using it due to ethical, factual, biased, and misinformation hazards [49]. All final claims, data interpretations, and conclusions remain the sole responsibility of the human authors. This implies that each author should be closely involved in the process: they must create questions or prompts for AI tools, assess the findings critically, and edit outputs based on their domain knowledge [50,51]. For example, a skilled researcher should provide AI with specific queries and keywords when conducting AI-assisted literature searches or creating hypotheses [52]. Likewise, when utilizing AI picture generators or graphics tools, writers must verify the uniqueness and accuracy of the images and properly credit any data sources. In summary, AI can assist

with drafting, but authorship still requires human creativity and judgment; it is unethical to use AI for a merely mechanical copy-paste job [9].

Learning to use AI skillfully is important. Authors should stay updated on best practices (e.g. prompt engineering) and avoid over-reliance [53]. For instance, generating hypotheses or treatment ideas with AI might be useful, but such suggestions must be validated through literature and expert consensus [54]. Training or guidance from institutions on using AI in medical writing can help people benefit from these tools while maintaining integrity [55].

6. Notes and disclosures for AI-assisted manuscripts

Given AI's prominent role, authors and journals must include explicit disclosures when AI tools are used [56]. Per updated ICMJE guidelines, authors must declare any use of AI technologies in both the cover letter and the submitted manuscript [9]. The tool name and version should be specified in the disclosure (e.g., "ChatGPT v4") and its purpose stated (e.g., language polishing, data analysis). A recommended format (as in vascular society guidelines) is a separate section [9,57]:

"During preparation of this work, the authors used [TOOL] to [reason]. The content was reviewed and edited by the authors, who take full responsibility for the manuscript." [9,57]

Such transparency ensures readers and editors know what content was AI-generated [58]. To uphold academic integrity, authors must transparently disclose AI use, verify outputs for accuracy and originality, and retain full responsibility—AI tools cannot be credited or listed as co-authors [59]. ICMJE explicitly requires authors to certify no plagiarism in any AI-produced text or images [9]. Reused pictures from AI-assisted design must be cited or marked as updated, and any quotes from AI outputs should be checked. Authors cannot conceal the usage of AI by identifying AI as an author or by deleting disclosures [9,60].

Standard ethics requirements—such as conflict-of-interest declarations and data transparency statements—remain applicable regardless of AI involvement [51,61]. The use of AI tools complements existing transparency obligations but does not replace established reporting standards.

Authors should also disclose any third-party editing or writing assistance: such agencies must be named in acknowledgments, and their contributions treated as minor (non-author) assistance [9,62,63]. Finally, all authors listed must fulfill the authorship criteria and collectively accept responsibility for the work, as per the ICMJE guidelines. This reaffirms the principle that human authorship entails accountability for content – AI or not [9,64].

7. Upgrading author guidelines and ethics statements

Medical journals must update their instructions for authors and editorial policies to reflect these issues [65]. Already, major editorial bodies have issued new guidance: for instance, the ICMJE revised its Recommendations (April 2025) to address AI, authorship, and transparency [9]. ICMJE now advises that the use of AI be disclosed and that AI not be listed as a legal author [9]. Similarly, COPE and WAME explicitly state that AI tools cannot meet authorship requirements and must be cited if used in writing [66,67]. Council of Science Editors (CSE) emphasize zero tolerance for plagiarism of any kind and urge clear ethics statements on reusing material [12,68].

In practice, journals should revise "Instructions for Authors" to mention AI and plagiarism explicitly [69]. This could mean demanding picture licenses, requiring a declaration of AI usage, and forbidding language recycling beyond basic allowances (with appropriate reference to past work) [70]. Ethics sections should remind authors that submitting a manuscript implies adherence to these rules, regardless of whether the material was human- or AI-generated [71]. A previous study found that ethics statements commonly address COI and consent but often omit specifics on AI and plagiarism [13]. We encourage

Ethical Area	Recommended Updates to Author Instructions
Plagiarism & Originality	Require authors to run manuscripts through originality-checking software (e.g., iThenticate) prior to submission. Prohibit text/image recycling.
AI Use in Writing	Authors must disclose the name, version, and purpose of any AI tools used (e.g., "ChatGPT v4, for grammar polishing"). AI cannot be listed as author.
Graphics & Figures	Authors must ensure figures are original or have documented permission. AI- or software-generated graphics must be acknowledged.
Authorship & Contributions	Reinforce that only individuals meeting authorship criteria (ICMJE) may be listed; professional editors/agencies must be acknowledged separately.
Conflict of Interest (COI)	Authors must disclose all financial and non-financial conflicts. Journals should require detailed COI statements at submission and publication.
Data Availability	Require Data Availability Statements in all original research and reviews, including datasets used by AI-based analyses.
Ethics Statements	Explicitly require disclosure of IRB/ethics approval for human/animal research and statement of informed consent where applicable.
Reviewer & Editor Guidance	Journals should update reviewer/editor instructions to recognize AI-generated content, similarity reports, and best practices for ethical evaluation.

Fig. 2. Recommendations are based on ICMJE and COPE guidance, supported by recent literature on AI and publication ethics [3,9,66,72].

journals to add clauses like “Authors must report any use of AI tools” and “Plagiarism of text or figures is unacceptable, even if adapted.” Uniform adoption of COPE’s updated positions will help standardize expectations across publications [66]. In Fig. 2 we found recommended updates to author instructions and ethics statements in medical journals, aligned with ICMJE and COPE guidance [9,66].

To improve practical implementation, journals may consider adopting a structured editorial framework:

1. Mandatory AI-Use Disclosure Statements
2. AI Similarity Screening Prior to Peer Review
3. Transparent Similarity Threshold Policies
4. Image Verification Workflows
5. Regular AI and Publication Ethics Training
6. Explicit Prohibition of Listing AI Tools as Authors

8. Perspectives and conclusions

The ethical landscape of medical publishing is rapidly evolving as traditional concerns—plagiarism, redundant publication, and authorship disputes—intersect with challenges introduced by AI. While text-matching software, such as iThenticate, helps detect verbatim copying, visual and idea plagiarism remain difficult to identify, emphasizing the need for ongoing author training and adherence to COPE and ICMJE guidelines [9,12,66]. Although large language models and AI-driven tools can improve writing quality and productivity, they also carry the potential for over-reliance, unrecognized use, and biased or false outputs, which necessitate strict human oversight and open disclosure [44,47–49]. Plagiarism and AI misuse both threaten scholarly integrity, and prevention requires a proactive approach that combines software detection, ethical training, mandatory AI disclosure, and explicit editorial policies [9,12,47]. Ultimately, maintaining trust in medical publishing depends on reaffirming that, while AI can assist, only human authors assume responsibility for accuracy, originality, and adherence to ethical standards [9,64,66].

In conclusion, it is necessary to strike a balance between the potential benefits and problems presented by AI and conventional ethical protections to maintain integrity in medical publication. Preventing

plagiarism, identifying the author, and maintaining complete transparency are still crucial, and AI techniques must be used sensibly, openly, and under the constant supervision of human writers. Ultimately, trust in scholarly work depends on the principle that technology can assist but cannot replace the accountability, judgment, and ethical responsibility of researchers.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Statement

During the preparation of this work, the authors used Grammarly (<https://app.grammarly.com/>) in order to enhance the grammar and clarity of this manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Data availability

No data was used for the research described in the article.

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